

Problem Set #13

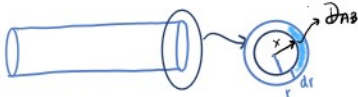
Sunday, April 19, 2020 2:14 AM

14.18, 14.24 and 14.33

14.18 Consider the radial diffusion of a gaseous species (A) through the wall of a plastic tube (B), and allow for chemical reactions that provide for the depletion of A at a rate $\dot{N}_A$ (kmol/s · m³). Derive a differential equation that governs the molar concentration of species A in the plastic. Compare your result with Equation 14.49.

Cylindrical Coordinates:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(CD_{AB} r \frac{\partial x_A}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \phi} \left(CD_{AB} \frac{\partial x_A}{\partial \phi} \right) + \frac{\partial}{\partial z} \left(CD_{AB} \frac{\partial x_A}{\partial z} \right) + \dot{N}_A = \frac{\partial C_A}{\partial t} \quad (14.49)$$



conducting a mass balance on this gets us:

$$\text{In} - \text{Out} + \text{Gen} - \text{Cons} = \text{Accum} \quad \text{Steady State} = 0$$

$$N_{A,r} - N_{A,r+dr} + N_{A,g} = 0$$

$$- (N_{A,r} + \frac{\partial N_{A,r}}{\partial r} dr)$$

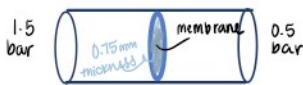
$$-\frac{\partial N_{A,r}}{\partial r} dr - N_A (2\pi r dr) = \frac{\partial [C_A (2\pi r dr)]}{\partial t}$$

$$-2\pi D_{AB} \frac{\partial}{\partial r} \left(r \frac{\partial C_A}{\partial r} \right) dr - N_A (2\pi r dr) = 2\pi r dr \frac{\partial C_A}{\partial t}$$

$$N_A = \frac{2\pi r dr \left(\frac{\partial C_A}{\partial t} \right) + 2\pi D_{AB} \frac{\partial}{\partial r} \left(r \frac{\partial C_A}{\partial r} \right)}{-2\pi r dr}$$

$$N_A = -\frac{\partial C_A}{\partial t} - \frac{D_{AB}}{r} \frac{\partial}{\partial r} \left(r \frac{\partial C_A}{\partial r} \right)$$

14.24 Oxygen gas is maintained at pressures of 1.5 bar and 0.5 bar on opposite sides of a rubber membrane that is 0.75 mm thick, and the entire system is at 25°C. What is the molar diffusive flux of O₂ through the membrane? What are the molar concentrations of O₂ on both sides of the membrane (outside the rubber)?



$$\text{diffusive flux: } J''_{A,x} = -D_{AB} \frac{dC_A}{dx}$$

$$= -D_{AB} \frac{C_A(0) - C_A(L)}{L}$$

L = 0.75 mm

TABLE A.10 The Solubility of Selected Gases and Solids*

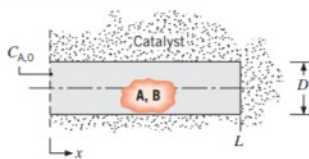
Gas	Solid	T (K)	S = C _A /p _A (kmol/m ³ · bar)
O ₂	Rubber	298	3.12 × 10 ⁻³
N ₂	Rubber	298	1.56 × 10 ⁻³
CO ₂	Rubber	298	40.15 × 10 ⁻³
He	SiO ₂	293	0.45 × 10 ⁻³
H ₂	Ni	358	9.01 × 10 ⁻³

*Data from Reference 26.

TABLE A.8 Binary Diffusion Coefficients at One Atmosphere*

Substance A	Substance B	T (K)	D _{AB} (m ² /s)
Gas			
He	Air	273	0.20 × 10 ⁻⁴
H ₂	Air	273	0.26 × 10 ⁻⁴
CO ₂	Air	273	0.14 × 10 ⁻⁴
H ₂	Air	298	0.21 × 10 ⁻⁴
O ₂	Air	298	0.21 × 10 ⁻⁴
Acetone	Air	298	0.10 × 10 ⁻⁴
Benzene	Air	298	0.09 × 10 ⁻⁴
N ₂	Air	298	0.18 × 10 ⁻⁴
CO ₂	Air	298	0.14 × 10 ⁻⁴
H ₂	N ₂	273	0.26 × 10 ⁻⁴
O ₂	N ₂	273	0.18 × 10 ⁻⁴
CO ₂	N ₂	273	0.14 × 10 ⁻⁴
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He	N ₂	273	0.20

- 14.33 To enhance the effective surface, and hence the chemical reaction rate, catalytic surfaces often take the form of porous solids. One such solid may be visualized as consisting of a large number of cylindrical pores, each of diameter D and length L .



Consider conditions involving a gaseous mixture of A and B for which species A is chemically consumed at the catalytic surface. The reaction is known to be first order, and the rate at which it occurs per unit area of the surface may be expressed as $k_1'' C_A$, where k_1'' (m/s) is the reaction rate constant and C_A (kmol/m³) is the local molar concentration of species A. Under steady-state conditions, flow over the porous solid is known to maintain a fixed value of the molar concentration $C_{A,0}$ at the pore mouth. Beginning from fundamentals, obtain the differential equation that governs the variation of C_A with distance x along the pore. Applying appropriate boundary conditions, solve the equation to obtain an expression for $C_A(x)$.

$$\text{In} - \text{Out} + \text{Gen} - \text{Consum} = 0$$

$$N_{A,x} - N_{A,x+dx} - k_1'' C_A = 0$$

$$- \frac{dN_A}{dx} dx - k_1'' C_A (\pi D) dx = 0$$

$$\left(-C_{DAB} \frac{dC_A}{dx} \right) \frac{\pi D^2}{4} - \frac{\pi D^2}{4} D_{AB} \frac{d^2 C_A}{dx^2} - k_1'' C_A (\pi D) dx = 0 \rightarrow \frac{d^2 C_A}{dx^2} - \frac{k_1'' C_A}{D_{AB}} = 0$$

non-homogeneous DE: $C_1 e^{mx} + C_2 e^{-mx} = y$ $m = \sqrt{\frac{k_1''}{D_{AB}}}$

BC#1 @ $x=0$, $C_A = C_{A,0}$

$$C_1 + C_2 = C_{A,0}$$

BC#2 $\frac{dC_A}{dx} \bigg|_{x=L} = \frac{k_1'' C_{A,L}}{D_{AB}}$

BC#1 $C_1 e^{my} + C_2 e^{-my} = y$ $C_1 e^{my} + (-C_1 + C_{A,0}) e^{-my} = \frac{k_1'' C_{A,L}}{D_{AB}}$

$$C_1 e^{my} + C_2 e^{-my} - C_{A,0} e^{-my} = \frac{k_1'' C_{A,L}}{D_{AB}}$$

$$C_1 (e^{my} + e^{-my}) = \frac{k_1'' C_{A,L}}{D_{AB}} + C_{A,0} e^{-my} \rightarrow C_1 = \frac{\left(\frac{k_1'' C_{A,L}}{D_{AB}} \right) + C_{A,0} e^{-my}}{m (e^{my} + e^{-my})} \therefore C_2 = C_{A,0} - \frac{\left(\frac{k_1'' C_{A,L}}{D_{AB}} \right) + C_{A,0} e^{-my}}{m (e^{my} + e^{-my})}$$

recall, that $m = \sqrt{\frac{k_1''}{D_{AB}}}$

$$y = \left[\frac{\left(\frac{k_1'' C_{A,L}}{D_{AB}} \right) + C_{A,0} \left(\sqrt{\frac{k_1''}{D_{AB}}} e^{-\sqrt{\frac{k_1''}{D_{AB}}} L} \right)}{\sqrt{\frac{k_1''}{D_{AB}}} \left(e^{\sqrt{\frac{k_1''}{D_{AB}}} L} + e^{-\sqrt{\frac{k_1''}{D_{AB}}} L} \right)} \right] e^{\sqrt{\frac{k_1''}{D_{AB}}} x} + \left[C_{A,0} - \frac{\left(\frac{k_1'' C_{A,L}}{D_{AB}} \right) + C_{A,0} \left(\sqrt{\frac{k_1''}{D_{AB}}} e^{-\sqrt{\frac{k_1''}{D_{AB}}} L} \right)}{\sqrt{\frac{k_1''}{D_{AB}}} \left(e^{\sqrt{\frac{k_1''}{D_{AB}}} L} + e^{-\sqrt{\frac{k_1''}{D_{AB}}} L} \right)} \right] e^{-\sqrt{\frac{k_1''}{D_{AB}}} x}$$